Cheatsheet AZ-305 — 10 · Backup, business continuity, DR and hybrid networking

1. Backup vs Site Recovery

	Azure Backup	Azure Site Recovery (ASR)
What it is	Backup copies in a Recovery Services vault	Continuous VM replication and orchestrated failover
Protects against	Corruption, deletion, ransomware, data loss	Datacenter / region outage (continuity)
RPO	Hours (backup frequency)	Minutes
RTO	Hours (restore)	Minutes (Recovery Plan)
Cost	Low	Low (no active VM in the target until failover)
When	"recover the data", "retain N years"	"remain available", "relaxed RPO + min cost", VM migration

- **SQL on VM, nightly batch, RPO 24 h, RTO 15 min, regional, automated, min cost → ASR** (a multi-region AG would be expensive; backup does not meet the RTO).
- **Changing vault:** a VM can only be in one vault → first **Stop backup** in the source vault.
- **The VM's Disaster recovery blade = ASR**, not Backup.

2. Azure Backup agents and components

Component	What it protects	When
VM backup extension	Entire Azure VMs	VM already in Azure; no agent to install
MARS agent	Files and folders on Windows servers (on-prem or Azure)	"specific files/folders", simplest solution
MABS (Azure Backup Server = DPM for Azure)	Full applications: SQL Server, Exchange, SharePoint, Hyper-V/VMware VMs, bare-metal	"application-aware", "centralized backup of many on-prem servers"
DPM (System Center)	Same as MABS, with a System Center license	They already have System Center
SQL backup on Azure VM	SQL databases with log backups every 15 min	RPO 15 min for SQL on VM
Backup of Azure Files / Blobs / Disks / PaaS DBs	PaaS resources	Depends on the resource

- Long retention (7 years, two separate locations) → vault with **GRS** + long-term retention; historical alternative DPM + tape.
- **Soft delete for backups** (14 days) and **Multi-user authorization with Resource Guard** against ransomware.

3. Resource Guard and roles

Element	Detail
Resource Guard	Resource in **another subscription/tenant**; critical vault operations (disable soft delete, reduce retention, stop backup) require a role on it → anti-ransomware separation of duties
Contributor	Create and manage the vault
Site Recovery Contributor	Enable replication, manage jobs (does not create the vault)
Site Recovery Operator	Run failover / failback
Backup Contributor / Operator / Reader	Equivalents for Backup

4. Storage: continuity

- **GRS/RA-GRS** for region; **ZRS** for datacenter; failover leaves the account in **LRS** → reconfigure GRS for failback.
- **Azure File Sync** for continuous file access when a site goes down (it is not backup).
- **Immutable storage** protects against deletion/modification, it does not restore.

5. Hybrid networking

Technology	Medium	Encryption	Bandwidth / SLA	When
Site-to-Site VPN	Internet	IPsec/IKE	Up to ~10 Gbps (SKU), no network SLA	Offices ↔ Azure; **ExpressRoute failover** ("must use Internet, no MPLS")
Point-to-Site VPN	Internet	SSTP/IKEv2/OpenVPN	Per client	Remote users; certificates (public root is uploaded, client stays) or Entra ID
ExpressRoute	Private circuit (provider, MPLS)	**Not encrypted by default**; add IPsec over ER or **MACsec** (ER Direct)	Up to 100 Gbps, SLA 99.95 %, **redundant pair of cross connections**	Private, predictable, high volume
ExpressRoute Global Reach	Microsoft backbone	—	—	Connect sites to each other via ER
VNet peering	Backbone	—	—	VNet ↔ VNet (regional/global); **not transitive**; "Use remote gateways" to share a gateway

Technology	Medium	Encryption	Bandwidth / SLA	When
Virtual WAN	Managed hubs	—	—	Multi-region, transitive, automatic P2S, ER (Standard)
Azure Bastion	—	TLS	—	RDP/SSH without public IP
NAT Gateway	—	—	—	Outbound Internet with a fixed IP

- **Gateway Subnet** required for VPN/ER gateways; Local Network Gateway represents the on-prem network.
- **Zone-redundant gateways** (AZ SKUs) for HA; VPN Gateway **active-active** for redundancy.
- **Subnet sizing:** Azure reserves **5 IPs**; /27 = 27 usable, /26 = 59, /25 = 123, /24 = 251. Never overlap with on-prem.
- **UDR with NVA/firewall as next hop** forces inspection; a single NVA = single point of failure → pair of NVAs + ILB or Azure Firewall.
- **DNS:** Private DNS Zones for multi-VNet and private endpoints; privatelink.* zone per service.
- **Private Endpoint** (private IP for the service, blocks public access) vs **Service Endpoint** (subnet → service over the backbone, public IP is kept).

6. Network security

Service	Layer	Filters by	When
NSG	4	IP, port, protocol, service tags, ASGs	Subnet / NIC
Application Security Groups	4	Group NICs for NSG rules	Simplify NSGs
Azure Firewall	4-7	IP, port, **FQDN**, tags, threat intel; Premium: TLS inspection, IDPS	Central hub, FQDN filtering
WAF (App Gateway / Front Door)	7	OWASP, bots, rate limit	Web apps
DDoS Protection	3-4	Volumetric	Public IPs
JIT VM access (Defender)	—	Opens management ports temporarily	RDP/SSH on demand
Network Watcher	—	Diagnostics (IP flow verify, next hop, flow logs)	"allowed or denied"

7. Disaster recovery per service (cross summary)

Service	DR mechanism with minimum downtime	Cheap mechanism (data)
VM	ASR	Backup
SQL on VM	Asynchronous AG / Distributed AG	Backup + log shipping
Azure SQL DB	Auto-failover group	Geo-restore
Managed Instance	Auto-failover group	Geo-restore
MySQL/PostgreSQL Flexible	Promoted read replica	Geo-redundant backup
Cosmos DB	Multi-region (writes or automatic failover)	Continuous backup
Storage	RA-GRS / account failover	GRS + restore
App Service / AKS	Two regions + Front Door / Traffic Manager	Redeploy from IaC
Key Vault	Automatically replicated to the paired region	—

8. Decision tree

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What must survive and with what RPO/RTO?
├─ Data only, hours of RTO, min cost ............. Backup (GRS vault, retention)
├─ VM available in another region, RPO minutes ... ASR + Recovery Plan
├─ Files/folders of a server ..................... MARS
├─ SQL/Exchange/Hyper-V on-prem centralized ....... MABS
├─ Backups resistant to a compromised admin ...... Resource Guard in another subscription
├─ Private on-prem connection, SLA, high volume .. ExpressRoute (+ IPsec if encryption is required)
├─ ExpressRoute failover over the Internet ....... Site-to-Site VPN
├─ New offices with no special requirements ...... Site-to-Site VPN
├─ VNet ↔ VNet ................................... Peering (Virtual WAN if transitivity/multi-region)
└─ Traffic inspected by FQDN ..................... Azure Firewall (secured hub) + UDRs
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9. Signals → answer

Wording	Answer
"RPO 24 hours, RTO 15 min, regional, automated, min cost" (SQL VM)	Azure Site Recovery
"use RSV2 to protect VM2 (in RSV1), first step"	Stop backup in RSV1
"files and folders", "simplest"	MARS
"SQL Server / Exchange / SharePoint on-prem"	MABS
"ransomware" + "role in Sub2 first"	Resource Guard
"create and manage the vault"	Contributor
"ExpressRoute fails, use Internet, no MPLS"	Site-to-Site VPN

Wording	Answer
"private, redundant pair of cross connections"	ExpressRoute
"offices in two cities, new subscription"	Site-to-Site VPN
"encryption required over ExpressRoute"	IPsec over ER / MACsec
"30 VMs, on-prem 172.16.0.0/16, S2S VPN"	192.168.0.0/24
"force traffic through firewall between VNets"	UDRs with the firewall as next hop
"packets allowed or denied"	Network Watcher
"RDP/SSH without public IP"	Bastion
"temporary management ports"	JIT VM access

10. Mistakes to avoid

1. Backup when an RTO of minutes is required → ASR.
2. Configuring RSV2 without first stopping the backup in RSV1.
3. Confusing the "Disaster recovery" section (ASR) with Backup.
4. Assuming ExpressRoute encrypts.
5. A second ExpressRoute as failover when MPLS is forbidden → VPN.
6. Forgetting the 5 reserved IPs when sizing subnets.
7. Treating peering as transitive.